

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs (ACJC – YJC)

1. [AJC/H2/Prelims 2018/P1/2]

A cannon at the top of a 30 m high hill fires a shell at an angle of 30° upwards from the horizontal with a speed of 50 m s^{-1} .

Taking air resistance to be negligible, what is the angle to the vertical at which the shell lands on level ground?

- A 39° B 42° C 48° D 51°

2. [CJC/H2/Prelims 2018/P1/4]

An object is thrown vertically upwards in air in which the air resistance is not to be neglected.

If the times of flight for the upward motion t_u and the time of flight to return to the same level t_d are compared, then

- A $t_d > t_u$, because the object moves faster on its downward flight and therefore the air resistance is greater.
 B $t_d = t_u$, because the effect of the air resistance is the same whether the object is moving upwards or downwards.
 C $t_d < t_u$, because at any given speed the net force when the object is moving downwards is greater than the net force when it is moving upwards.
 D $t_d > t_u$, because at a given speed the net force when the object is moving downwards is smaller than the net force when it is moving upwards.

3. [DHS/H2/Prelims 2018/P1/3]

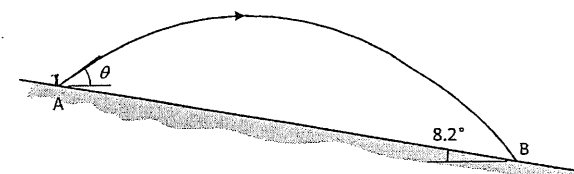
A projectile of mass 2.0 kg is launched on the Earth with some initial velocity. Another projectile of mass 4.0 kg is launched on the Moon with the same initial velocity. The acceleration of free fall on the Moon is 1.6 m s^{-2} .

Neglecting air resistance, what is the ratio $\frac{\text{range of projectile on the Earth}}{\text{range of projectile on the Moon}}$?

- A 0.16 B 0.33 C 3.3 D 6.1

4. [DHS/H2/Prelims 2018/P1/4]

A golf ball is hit from point A on the ground and moves through the air to point B as shown in the figure which is not drawn to scale.

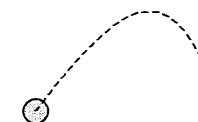


The ground slopes downhill with constant gradient of angle 8.2° to the horizontal. The ball has an initial velocity of 63 m s^{-1} at an angle θ to the horizontal. Time taken for the ball to travel from A to B is 4.9 s. Determine the angle θ .

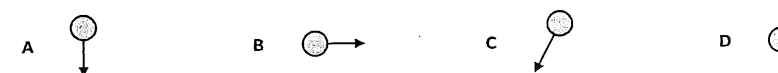
- A 8.0° B 10° C 12° D 14°

5. [EJC/H2/Prelims 2018/P1/3]

A ball thrown at an angle travels in a trajectory as shown below.

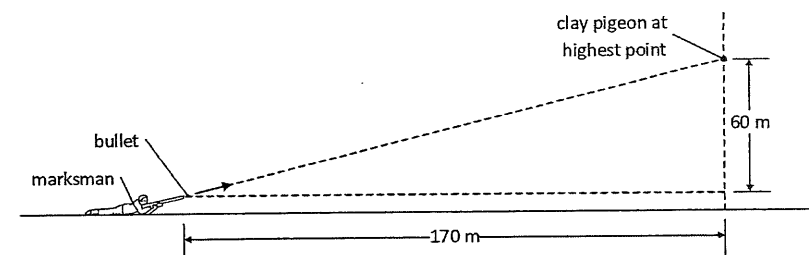


When the ball is at the top of its flight, which of the following shows the direction of its resultant acceleration?



6. [HCI/H2/Prelims 2018/P1/3 Modified]

A clay pigeon is launched vertically into the air from the ground.



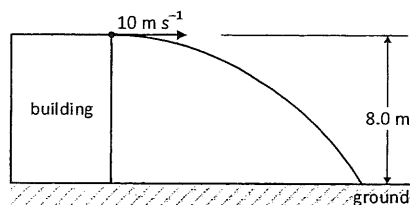
A marksman lies at a horizontal distance of 170 m away from the launching device. When the clay pigeon reaches its maximum height of 60 m above the muzzle, the marksman aims his rifle at the clay pigeon and fires a bullet at it. The bullet leaves the rifle with a speed of 300 m s^{-1} .

At which time after the bullet is fired, does the bullet hit the clay pigeon? Assume air resistance is negligible.

- A 0.17 s B 0.57 s C 0.60 s D 1.66 s

7. [IJC/H2/Prelims 2018/P1/3]

A boy throws a stone with a horizontal velocity of 10 m s^{-1} from the top of a building. The height of the building is 8.0 m . The stone travels along a curved path until it hits the ground, as shown in the diagram.

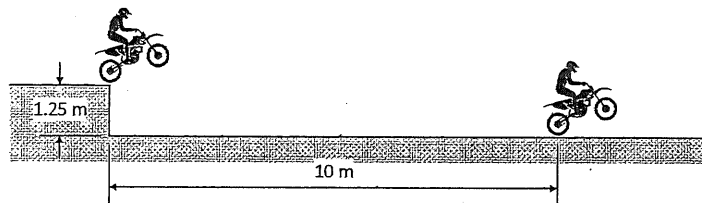


Neglecting air resistance, how long does it take the stone to reach the ground?

- A 0.61 s B 0.80 s C 1.3 s D 1.6 s

8. [MJC/H2/Prelims 2018/P1/3]

A motorcycle stunt-rider moving horizontally takes off from a point 1.25 m above the ground, landing 10 m away as shown in the diagram.



What was the speed at take-off?

- A 5 m s^{-1} B 10 m s^{-1} C 15 m s^{-1} D 20 m s^{-1}

9. [RVHS/H2/Prelims 2018/P1/3]

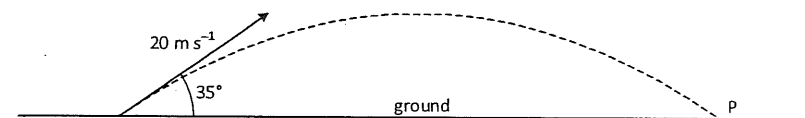
A ball is thrown vertically upwards. It reaches its maximum height of motion and comes back to the original point where it was launched.

Given that air resistance to the motion of the ball is not negligible, which of the following statements about the ball is correct?

- A The ball is accelerating at 9.81 m s^{-2} throughout its motion.
 B The ball is decelerating at a decreasing rate when it is moving upwards.
 C The ball is decelerating at an increasing rate when it is moving upwards.
 D The ball is accelerating at an increasing rate when it is moving downwards.

10. [TPJC/H2/Prelims 2018/P1/3]

A student throws a stone 35° above the horizontal at an initial speed of 20 m s^{-1} . It travels in projectile motion until it hits the ground at P with the same speed.

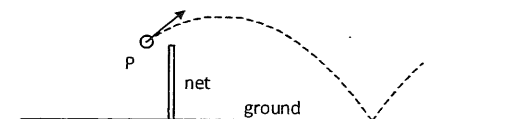


What is the magnitude of the change in velocity of the stone just before hitting the ground at P?

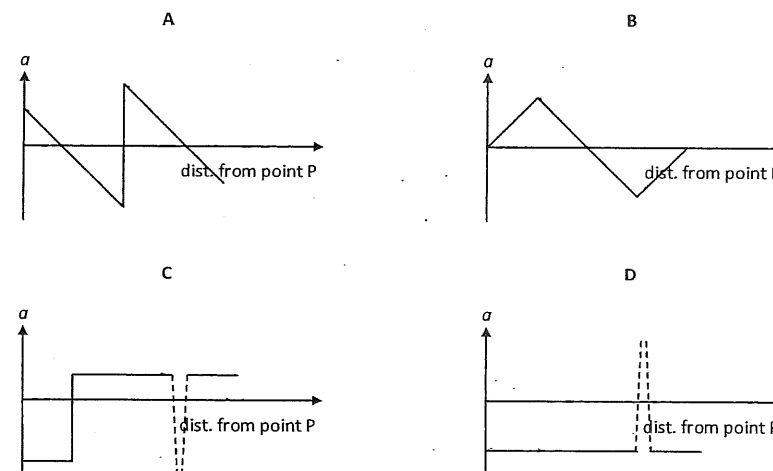
- A 0 m s^{-1} B 20 m s^{-1} C 23 m s^{-1} D 33 m s^{-1}

11. [TPJC/H2/Prelims 2018/P1/4]

The diagram below shows the trajectory of a tennis ball crossing the net and bouncing once from the ground.



Which of the following graphs represents the variation with the horizontal distance from point P of the acceleration a of the ball, taking the upward direction as positive?



12. [YJC/H2/Prelims 2018/P1/2]

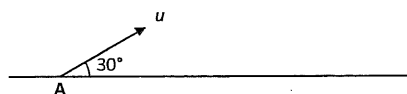
A ball is thrown horizontally from the top of a tower and fall towards the ground below. The initial velocity of the stone is u .

Assuming that air resistance is negligible, what is the vertical velocity and horizontal displacement of the ball t seconds after leaving the tower?

	Vertical velocity	Horizontal displacement
A	gt	$\frac{1}{2}gt^2$
B	$u+gt$	$ut + \frac{1}{2}gt^2$
C	gt	$ut + \frac{1}{2}gt^2$
D	gt	ut

13. [HCI/H2/Prelims 2018/P1/4]

A projectile of mass m is fired at ground level with velocity u from a point A, as shown below.

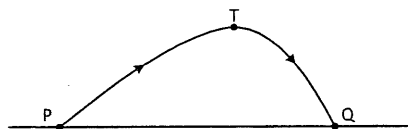


Neglecting air resistance, determine the magnitude of the change in momentum of the mass between leaving point A and arriving back at ground level.

- A zero B $\frac{1}{2}mu$ C mu D $2mu$

14. [JJC/H2/Prelims 2018/P1/2]

In the presence of air resistance, a stone is thrown from P and follows a path in which the highest point reached is T as shown in the diagram below.



Given that the drag force acting on the stone is directly proportional to the magnitude of its instantaneous velocity, the vertical component of the acceleration of the stone is

- A highest at point P.
B highest at point T.
C highest at point Q.
D constant through the travelled path.

II. Structured Questions (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P2/2]

At $t = 0.0 \text{ s}$, a golf ball is launched with initial velocity u and angle θ to the horizontal as shown in Fig. 1.1. After 7.0 s , it lands at a distance of 220 m away at the same level where it was launched.

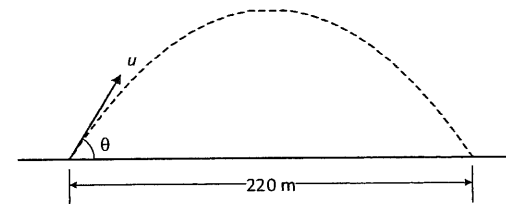


Fig. 1.1

- a. Determine u and θ . [5]
b. i. Calculate magnitude of the change in velocity, Δv , of the golf ball in the duration $t = 0.0 \text{ s}$ to $t = 7.0 \text{ s}$. [1]
ii. By scale drawing, draw the vectors representing u and Δv . Hence, determine the magnitude and direction of the velocity v of the ball at 7.0 s . Label the vectors clearly. [3]

[Total: 9]

2. [MJC/H2/Prelims 2018/P3/4]

A person threw a ball vertically upwards.

- a. Fig. 2.1 shows the variation with time t of the velocity v when air resistance is absent.

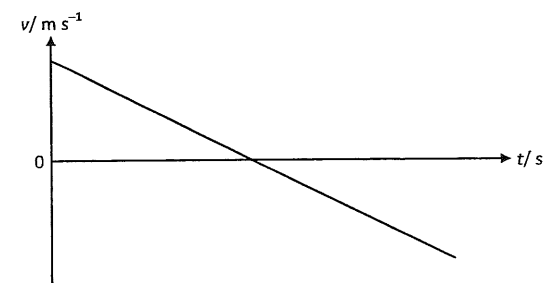


Fig. 2.1

Draw on Fig. 2.1 a second graph for the case where air resistance is present.

- b. Explain how the presence of air resistance would affect the maximum height reached by the ball.

[Total: 4]

3. [NJC/H2/Prelims 2018/P2/1]

A disc moves on a smooth horizontal surface. Fig. 3.1 shows the top view of the disc moving from point P at $t = 0.0$ s with an initial velocity of 0.200 m s^{-1} in the north-east direction. From $t = 0.0$ s to $t = 10.0$ s, the disc has a constant acceleration of 0.100 m s^{-2} in the easterly direction.

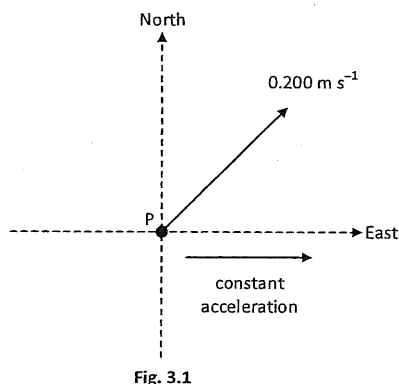


Fig. 3.1

- Determine the disc's velocity at $t = 10.0$ s. [3]
- Determine the distance of the disc from the point P at $t = 10.0$ s. [3]
- From $t = 10.0$ s to $t = 20.0$ s, the acceleration of the disc has a constant value of 0.100 m s^{-2} directed towards the west.
 - Sketch the velocity-time graph along the East-West direction between $t = 0.0$ s to $t = 20.0$ s. Take the eastward direction as positive. [2]
 - Sketch in Fig. 3.2 the path of the disc from $t = 0.0$ s to $t = 20.0$ s. [2]

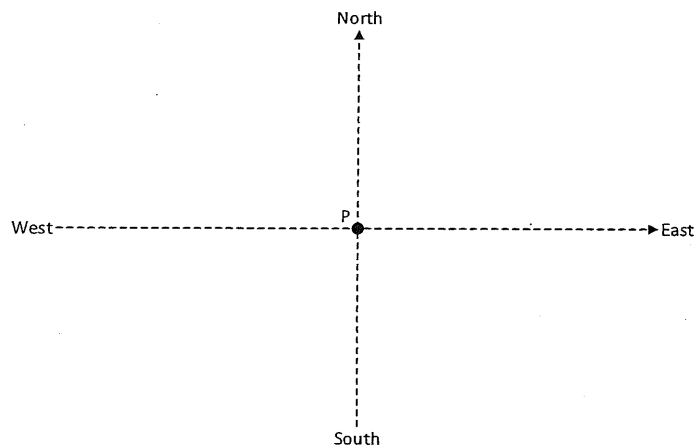


Fig. 3.2

[Total: 10]

4. [NYJC/H2/Prelims 2018/P2/2]

A grenade is launched at ground level with velocity of 75 m s^{-1} and angle 20° from the horizontal to hit a sniper on top of a 30 m tall building at a position 245 m from the foot of the building. The sniper is standing at a distance x from the edge of the building.

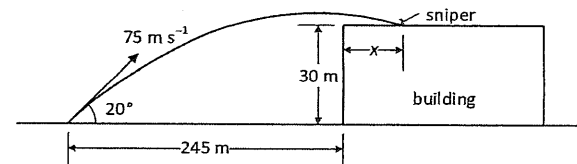


Fig. 4.1

- Show that the time for the grenade to reach the sniper is 3.5 s. [1]
- Calculate the speed of the grenade just before impact. [2]
- Determine the distance, x , of the sniper from the edge of the building. [2]
- The sniper runs away immediately from the edge of the building when the grenade is launched. Given that the grenade has a 'kill radius' of 10 m, calculate the minimum constant acceleration at which the sniper should run in order to escape. [1]

[Total: 6]

5. [PJC/H2/Prelims 2018/P3/1]

A ball is thrown from point O as shown in Fig. 5.1.

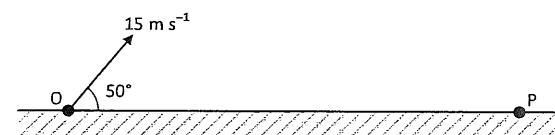


Fig. 5.1

The initial velocity of the ball is 15 m s^{-1} at an angle of 50° above the horizontal. The ball lands at point P. The points O and P are at the same horizontal level. Assume that air resistance is negligible.

- Determine the length of OP. [3]
- On Fig. 5.1, sketch the paths of the ball for the same given initial velocity
 - when there is no air resistance and label it A, [1]
 - when air resistance cannot be neglected and label it B. [2]

[Total: 6]

6. [SRJC/H2/Prelims 2018/P2/1]

A raindrop falls vertically from rest.

- Assume that air resistance is negligible.

On Fig. 6.1, sketch a graph to show the variation with time t of the velocity v of the raindrop for the first 1.0 s of the motion. [1]

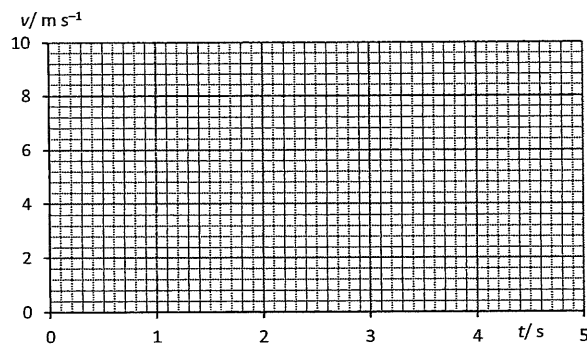


Fig. 6.1

- b. In practice, air resistance D on the raindrop is not negligible. D is given by the expression

$$D = kv^2$$

where k is a constant and v is the speed.

- i. The raindrop has mass 1.38×10^{-5} kg and k is 2.76×10^{-6} N m⁻² s².

Calculate the terminal velocity of the raindrop.

[2]

- ii. The raindrop reaches terminal velocity at $t = 5.0$ s.

On Fig. 6.1, sketch the variation with time t of velocity v for the raindrop. The sketch should include the first 5.0 seconds of the motion.

[2]

- c. A raindrop falls on a roof and rebounds off with a velocity of 5.5 m s⁻¹ at an angle 60° with respect to the horizontal as shown in Fig. 6.2.

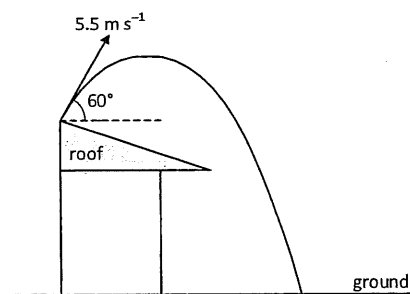


Fig. 6.2

Assume air resistance is negligible. The maximum horizontal distance travelled by the raindrop is 3.8 m.

- Calculate the time taken for the raindrop to hit the ground. [1]
- Determine the speed of the raindrop as it hits the ground. [2]
- Discuss quantitatively whether the assumption that air resistance is negligible is justified by considering the vertical component of the initial velocity of the raindrop. [2]

The raindrop has the same mass and dimension as in b.i.

[Total: 10]

7. [VJC/H2/Prelims 2018/P3/1 Part]

- Explain why it is technically incorrect to define speed as distance travelled per second. Include in your answer the correct statement defining speed. [2]
- An archer fires an arrow with an initial velocity u and hits a target which is a distance d away and on the same horizontal level as the bow, as illustrated in Fig. 7.1. The arrow is aimed so that, on release, it makes an angle θ with the horizontal.

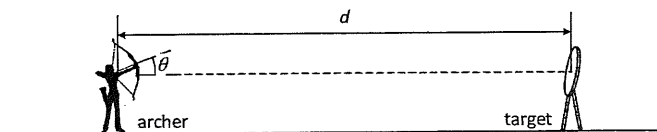


Fig. 7.1

- Assuming air resistance is negligible, write down an expression for d in terms of u , θ , and the time of flight t . [1]
- Write down an expression for the time of flight t in terms of u , θ and the acceleration of free fall g . [1]
- The distance d is given by the expression

$$d = \frac{u^2 \sin 2\theta}{g}$$

Calculate two possible angles, θ_1 and θ_2 , for an arrow with initial velocity $u = 32$ m s⁻¹ and a target at a distance d of 94 m from the bow. [3]

[Total: 7]

8. [TPJC/H2/Prelims 2018/P2/1 Part]

- A body is released in a fluid. With the aid of a free body diagram, explain how the body falling through a fluid can reach terminal velocity. [5]

9. [MI/H2/Prelims 2018/P3/7 Modified]

- a. A ball of mass 320 g is projected at an angle by a ball launcher. Air resistance is not negligible. The ball is launched from a height of 0.50 m above the level ground.

The variation with time t of the vertical velocity component V_y of the ball is shown in Fig. 9.1.

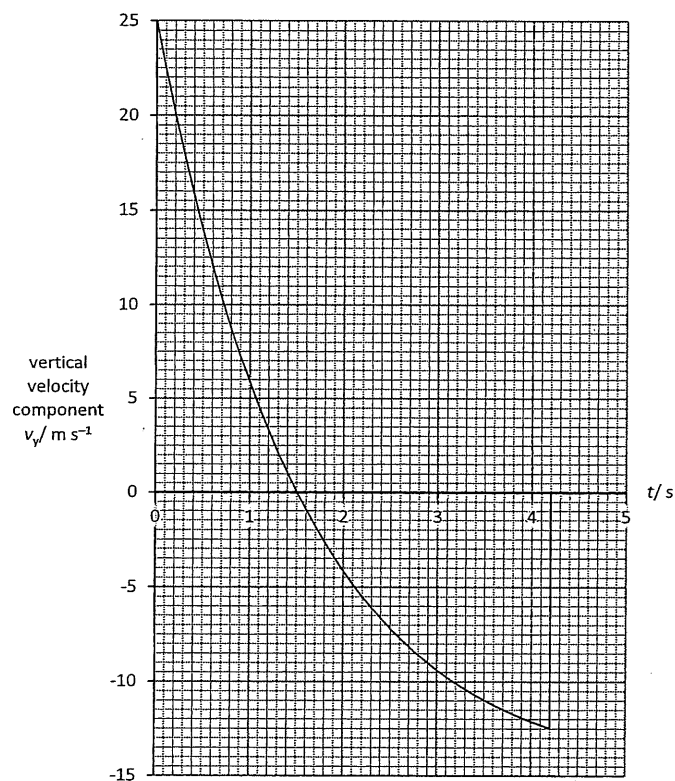


Fig. 9.1

- i. Estimate the vertical height from level ground reached by the ball at $t = 1.0$ s. [2]
- ii. Estimate the vertical component of the air resistance that is acting on the ball at $t = 1.0$ s. [3]
- iii. In the case that air resistance is negligible, the ball has a horizontal velocity component of 12.0 m s^{-1} . [1]
 1. On Fig. 9.1, sketch a graph to show the variation with time t of the velocity vertical component of the ball. Label this line Z. [2]
 2. Hence, determine the resultant velocity of the ball at $t = 1.0$ s. [2]

[Total: 10]